

AQA A-Level Mathematics

Paper 3 (7357/3) – Pure Maths & Statistics

Last-Minute Rescue Sheet | June 2026

This isn't a textbook. It's a targeted breakdown of what actually costs students marks on Paper 3 – built from examiner reports (2019–2024), mark schemes, and the exact phrasing examiners reward. Every section answers one question: where do marks get lost, and how do you stop the bleeding?

Two rules before you start: (1) If something is in the formula booklet, don't waste revision time memorising it. (2) Examiners don't care how clever your method is – they care whether you followed their script. Play their game.

1. Formula Triage

Stop memorising things the booklet already gives you. That's dead revision time. The left column is handed to you in the exam. The right column? If it's not in your head by Thursday morning, you're throwing marks away.

Pure Mathematics

Topic	IN THE BOOKLET (don't memorise)	NOT IN BOOKLET (must know cold)
Trig	$\sin(A \pm B)$, $\cos(A \pm B)$, $\tan(A \pm B)$ Small angle approx	$\sin^2\theta + \cos^2\theta = 1$ $\sec^2\theta = 1 + \tan^2\theta$ Exact values ($\sin 30^\circ$, $\cos 45^\circ$ etc.) Double angle formulas
Differentiation	Quotient Rule First principles limit formula	Product Rule: $(uv)' = u'v + uv'$ Chain Rule: $dy/dx = dy/du \times du/dx$ $d/dx(e^x) = e^x$, $d/dx(\ln x) = 1/x$ $d/dx(\sin x) = \cos x$, $d/dx(\cos x) = -\sin x$
Integration	Integration by Parts Trapezium Rule	$\int x^n dx = x^{n+1}/(n+1) + C$ $\int e^x dx = e^x + C$ $\int 1/x dx = \ln x + C$ $\int \sin x = -\cos x + C$
Numerical	Newton-Raphson formula Trapezium Rule	WHY N-R fails ($f'(x) = 0$ means division by zero – the tangent is horizontal)
Logs	–	$\ln(e^x) = x$, $e^{\ln x} = x$ Change of base: $\log_a(b) = \ln b / \ln a$ Log laws: add, subtract, power
Series	Quadratic formula Arithmetic/Geometric series sums	$(a+b)^n$ expansion formula Conditions for convergence of $(1+x)^n$

Statistics

Topic	IN THE BOOKLET	NOT IN BOOKLET (must know)
-------	----------------	----------------------------

Probability	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ Conditional: $P(A B) = P(A \cap B) / P(B)$ PMCC formula	Mutually exclusive: $P(A \cap B) = 0$ Independent: $P(A \cap B) = P(A) \times P(B)$
Distributions	Binomial formula: $P(X=r)$ Mean: np , Variance: $np(1-p)$ Normal standardisation: $Z = (X - \mu) / \sigma$	Conditions for Binomial: Fixed n , constant p , independent trials, two outcomes $E(X) = \sum x \cdot P(X=x)$
Data	Standard deviation / variance formulas (S_{xx})	Outlier boundaries: $Q1 - 1.5 \times IQR$ or $\mu \pm 2\sigma$ Frequency density = Frequency / Class width Linear interpolation formula

2. The Hypothesis Testing Blueprint

This is the highest-mark-density topic on Paper 3. Get the structure right and you're picking up 6–7 marks in under 10 minutes. Get it wrong – use the wrong word, skip a step, forget to define the parameter – and you'll drop 3–4 marks on a question you technically knew how to do. Here's the exact layout examiners want to see.

Worked Example

A machine fills pipes to a mean length of 50 cm. A new operator takes over. A sample of 30 pipes has a mean length of 50.8 cm. Lengths are normally distributed with standard deviation 2 cm. Test at 5% significance whether the mean length has increased.

Step 1 – Define the Parameter	Let μ be the true population mean length of the pipes (in cm).	<i>This is a free mark that about 40% of students throw away. Just write it.</i>
Step 2 – State Hypotheses	$H_0: \mu = 50$ $H_1: \mu > 50$	<i>Must use μ for a Normal mean test, p for Binomial. Never use x here.</i>
Step 3 – Distribution Under H_0	Under H_0 : $X \sim N(50, 2^2/30)$	<i>Don't forget to divide variance by n. Trips up a huge number of students.</i>
Step 4 – Calculate P-Value	$P(X \geq 50.8) = 0.0142$	<i>Use your calculator. Show the probability. Don't just say 'significant'.</i>
Step 5 – Explicit Comparison	$0.0142 < 0.05$	<i>Write this inequality down. Examiners specifically look for it.</i>
Step 6 – Conclusion in Context	Reject H_0 . There is sufficient evidence at the 5% significance level to suggest that the mean length of the pipes has increased.	<i>Three elements: 'sufficient/insufficient' + significance level + context from the question.</i>

Conclusion Language – What Scores vs. What Doesn't

SCORES MARKS	LOSES MARKS
"sufficient evidence"	"enough evidence"
"suggest that..."	"show" or "prove that..."
"reject H_0 " or "do not reject H_0 "	"accept H_1 " or only referring to H_1
"at the 5% significance level"	No mention of significance level

Conclusion references the actual scenario

Generic conclusion with no context

3. Statistics Decision Framework

The number one mistake on distribution questions isn't the maths – it's picking the wrong distribution in the first place. Here's how to tell them apart in under 10 seconds.

Distribution	When to Use	Conditions (all must hold)	Keyword Triggers
Binomial $X \sim B(n, p)$	Counting successes in a fixed number of trials	- Fixed number of trials n - Constant probability p - Trials are independent - Only two outcomes	"out of 20 items..." "probability of success is..." "how many pass/fail"
Normal $X \sim N(\mu, \sigma^2)$	Continuous measurements (height, weight, time, length)	- Continuous variable - Symmetric distribution - Mean and s.d. given	"normally distributed" "mean of... standard deviation of..."
Normal Approx to Binomial	When n is large and you need to approximate Binomial	$np \geq 5$ AND $n(1-p) \geq 5$ Continuity correction REQUIRED	Large sample + proportion test. $n = 200$ and $p = 0.3$ is your cue.

Continuity correction quick rule: Only use it when approximating a discrete distribution (Binomial) with a continuous one (Normal). If the data is already continuous, don't touch it.

A Note on the AQA Large Data Set

AQA uses the Department for Transport Vehicle Database – NOT the weather data set (that's Edexcel). Don't mix them up.

Feature	What You Need to Know
Scope	Cars registered in specific weeks in 2002 and 2016 only
Regions	London, North West, South West. No Scotland, no Wales.
Propulsion	PropulsionTypeId: 1 = Petrol, 2 = Diesel
Mass	Includes a standard 75 kg driver in the figure
Particulates	Only applies to diesel cars. Blank for petrol.
Makes	Five only: BMW, Ford, Toyota, Vauxhall, Volkswagen

4. Pure Maths – High-Frequency Topics & Trap Questions

Paper 3 tests Pure alongside Statistics. You can't coast on stats alone. These are the topics that show up most and the specific ways examiners try to catch you out.

Topic	What They'll Ask	The Trap / How Students Lose Marks
Proof	Direct proof, contradiction, exhaustion. Factor theorem proofs.	Proof by exhaustion: forgetting to test 0 or negatives. Contradiction: not stating the exact opposite of the

		premise. Factor theorem: not setting $f(-k) = 0$ properly.
Differentiation	Chain/product/quotient rule. Stationary points. Second derivative.	Finding the turning point but not determining its nature (max/min). Use $f''(x)$ or sign-change - pick one and commit.
Integration	By substitution, by parts. Differential equations.	The classic: substitution but forgetting to change limits from x to u . Diff equations: separating variables then forgetting $+ C$. That one mistake costs 2–3 marks.
Trig / Radians	Sector area, arc length. Identities in context.	Using degrees instead of radians. Calculator in wrong mode. Using θ when the angle is actually $(\pi - \theta)$.
"Show That"	Any topic. You're given the answer and asked to reach it.	Jumping steps. Using the printed answer within your working (circular reasoning). Not writing a final line stating the result matches.
Newton-Raphs on	"Explain why N-R fails for this starting value."	Don't calculate. The answer: " $f'(x_0) = 0$, so the tangent is horizontal, and the formula divides by zero." Done.

5. Mark Scheme Language

AQA examiners use a rigid system. M marks (method) reward the correct approach – even if your arithmetic is wrong. A marks (accuracy) reward the correct final answer. Here's the pattern that keeps showing up: students get M1 but lose A1. That's a solvable problem.

Scenario	M1 Awarded For	A1 Lost Because	How to Fix It
Any calculation	Correct method set up	Rounding intermediate values too early	Keep exact values in calculator until the final step.
Integration	Correct integral structure	Sign error or wrong power	Write the integral with numbers substituted BEFORE evaluating.
Hypothesis test	Correct setup and probability	Using "enough" not "sufficient"	Memorise the exact phrase template from Section 2.
Normal dist.	Correct standardisation	Sign error on z-value or wrong tail	Draw a quick sketch. Shade the area you want. 10 seconds saves marks.
Exact values	Correct method shown	Giving decimal when question says "exact value"	If it says exact, leave surds and fractions. Don't convert.

Golden rule from AQA's own marking guidance: "It is not possible to award M0 A1."

Translation: if you get the right answer but showed no method, you get zero. Always show your working.

6. The Last 30 Minutes – M1 Mark Grabbing

Clock's running down. Four questions left. You can't finish them all properly. Stop trying. Switch to harvest mode: grab method marks from every question rather than perfecting one.

Statistics – Free Marks

Action	Time Cost	Marks Gained
Write H_0 and H_1 with correct parameter (p or μ)	15 seconds	1–2 marks
Define the distribution: $X \sim B(n, p)$ or $X \sim N(\mu, \sigma^2)$	10 seconds	1 mark
Draw a Venn diagram or tree diagram for probability questions	1 minute	1–2 marks
Write the parameter definition ("Let p be the proportion of...")	10 seconds	1 mark

Pure Maths – Free Marks

Action	Time Cost	Marks Gained
Write out the formula with your numbers substituted in (don't just use calculator)	30 seconds	1 mark (M1)
For turning points: write $dy/dx = 0$ and attempt to differentiate	1 minute	1–2 marks
For integration: write the integral sign with limits. Even unsolved, it's a method.	20 seconds	1 mark (M1)
Indefinite integral? Write $+ C$ immediately. It's a mark on its own.	2 seconds	1 mark
Abandon messy algebra. The final A1 isn't worth 5 mins when you can grab 3 M1s elsewhere.	–	Net gain: 2–3

One last thing. Examiners aren't trying to trick you. They're testing whether you can communicate maths precisely. Show your method. Use the right words. Don't round early. That's 80% of the battle right there.

Built by StudyHours.com from AQA examiner reports (2019–2024), official mark schemes, and the AQA 7357 specification.